

TASK 11

Explore the java classes related to Digital Certificates.

Aim: To explore the Java classes related to Digital Certificates.

Program:

```
import java.io.*;
import java.security.*;
import java.util.*;

public class Task11 {
    private static Scanner s = new Scanner(System.in);
    public static void main(String[] args) {
        try {

            // Generating key pair (public and private keys)
            KeyPairGenerator keyGen = KeyPairGenerator.getInstance("DSA");
            keyGen.initialize(1024);
            KeyPair keyPair = keyGen.generateKeyPair();
            PrivateKey privateKey = keyPair.getPrivate();
            PublicKey publicKey = keyPair.getPublic();

            // Sample data to be signed
            String data = s.nextLine();
            byte[] dataBytes = data.getBytes();

            // Signing the data
            Signature signature = Signature.getInstance("SHA256withDSA");
            signature.initSign(privateKey); // Initialize for signing
            signature.update(dataBytes); // Add data
            byte[] digitalSignature = signature.sign(); // Generate the digital signature

            System.out.println("DigitalSignature:" + Base64.getEncoder().encodeToString(digitalSignature));

            // Verifying the signature
            signature.initVerify(publicKey); // Initialize for verification
            signature.update(dataBytes); // Add data to be verified
            boolean isVerified = signature.verify(digitalSignature); // Verify the signature

            System.out.println("Signature Verified: " + isVerified);
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

- 1.What is a **Public Key Infrastructure (PKI)**?
2. How does a digital certificate verify the authenticity of a user or website?
3. What are the key components of a digital certificate?
4. What is the difference between a self-signed certificate and a CA-issued certificate?